

Windows Server 2025 Active Directory Domain Services Deployment

Concise portfolio report covering deployment, configuration and verification

PREPARED BY

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Document purpose: Portfolio demonstration of Windows Server 2025 AD DS deployment, configuration, and verification.

Report Overview

This report summarizes the deployment of a **Windows Server 2025** domain controller for the isolated **mypersonal.ian** lab. It focuses on the decisions and evidence needed to confirm that **Active Directory Domain Services**, **DNS**, reverse **DNS**, **SYSVOL**, **NETLOGON**, and domain-controller discovery were working as expected.

Project outcome

WIN2K25-DC03 was configured as the first **domain controller** for the new forest. The server used a **static IPv4 address**, hosted the **DNS** service, exposed the expected domain shares, and returned successful forward, reverse, and **LDAP SRV** lookup results.

Environment summary

Item	Configuration
Operating system	Windows Server 2025
Server name	WIN2K25-DC03
Domain	mypersonal.ian
IPv4 address	172.16.131.21/24
Default gateway	172.16.131.2
DNS server	172.16.131.21
Installed roles	AD DS and DNS Server

1 Server Preparation

A stable hostname and **static network configuration** were established before installing Active Directory. The server used **172.16.131.21/24**, the default gateway **172.16.131.2**, and its own address for **DNS**. This prevents the address from changing and supports reliable **DNS** registration.

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 172 . 16 . 131 . 21

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 172 . 16 . 131 . 2

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 172 . 16 . 131 . 21

Alternate DNS server: | . . . |

☐ Validate settings upon exit

Advanced...

OK Cancel

Figure 1. Static IPv4 and DNS settings assigned to WIN2K25-DC03.

2 Installing Active Directory Domain Services

The **Active Directory Domain Services** role was selected. The installation completed before the server was promoted to a **domain controller**.

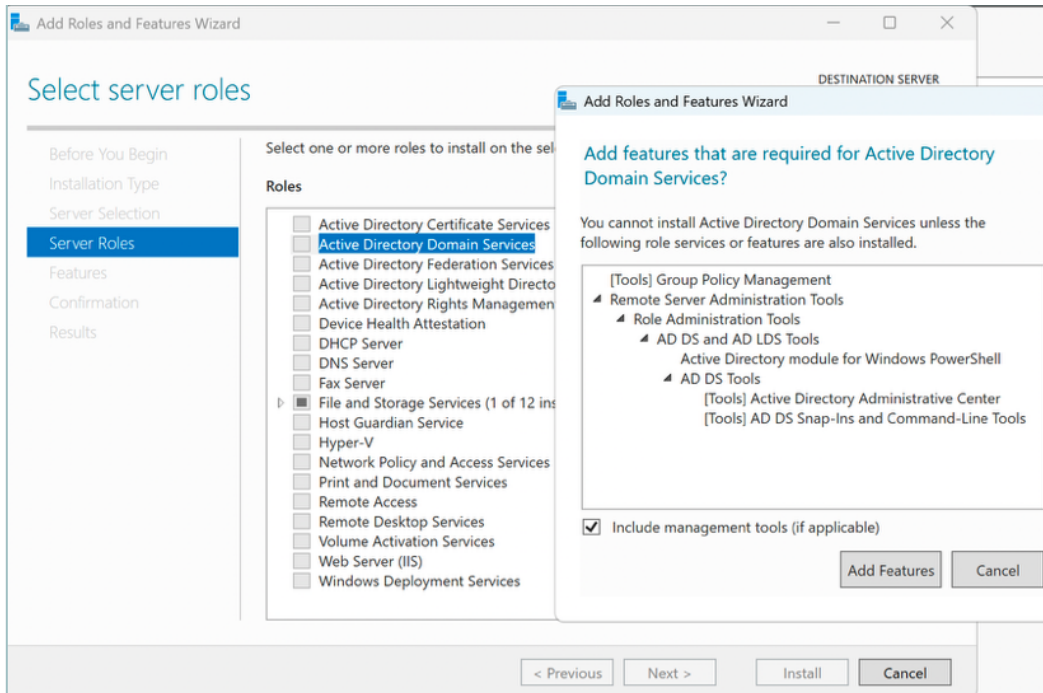


Figure 2. Active Directory Domain Services selected on the Server Roles page.

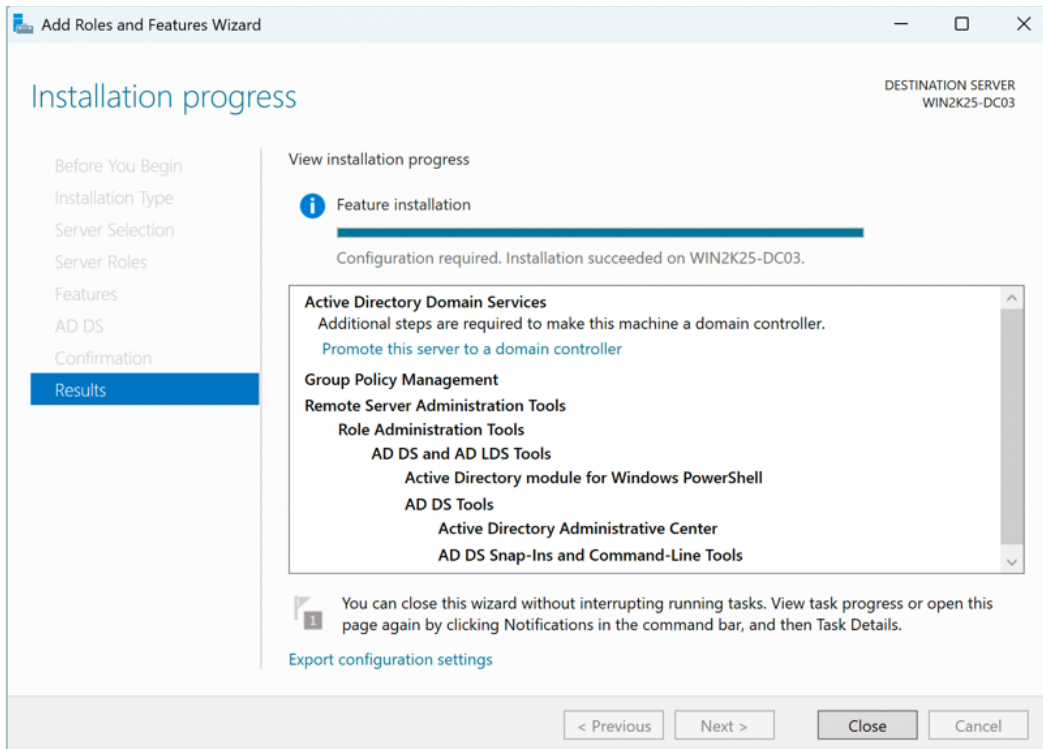


Figure 3. AD DS role installation completed successfully.

3 Creating the New Forest

The server was configured as the first **domain controller** in a new forest named **mypersonal.ian**. **DNS Server** and **Global Catalog** were enabled, and a **Directory Services Restore Mode** password was entered for recovery use.

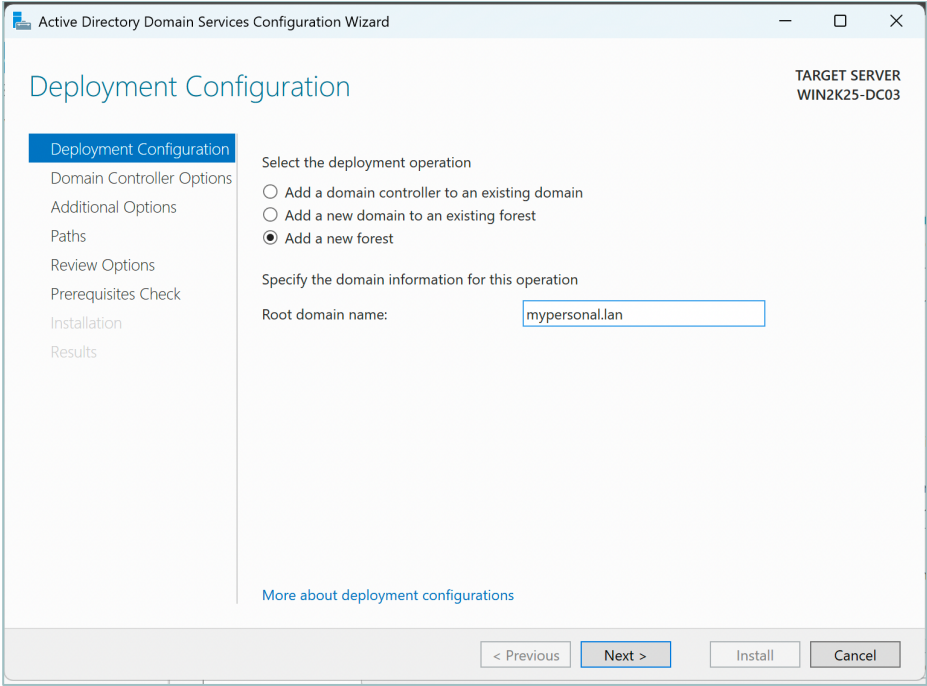


Figure 4. New forest selected with mypersonal.ian as the root domain.

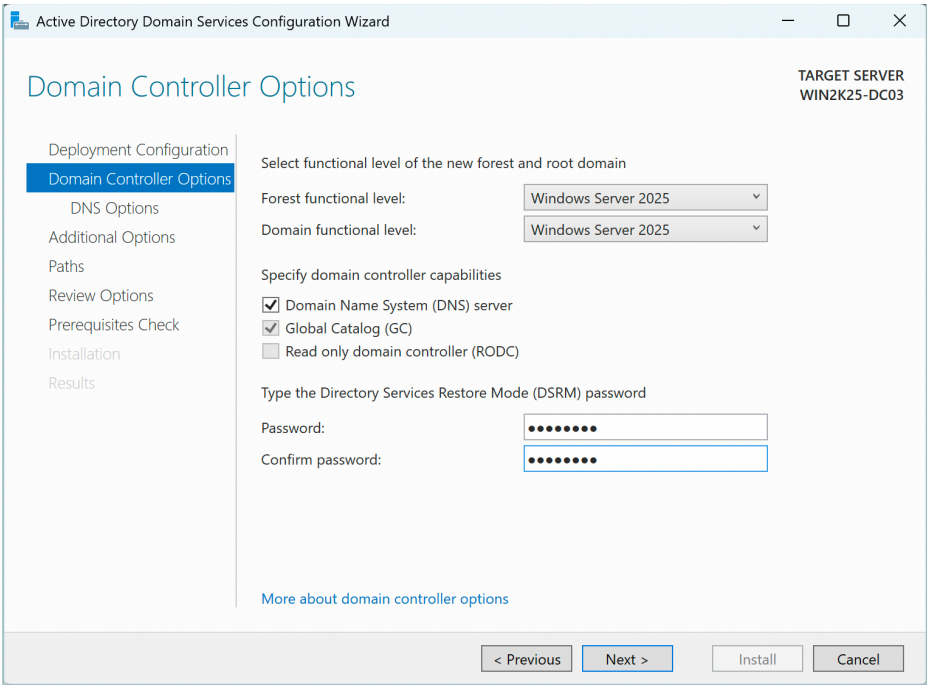


Figure 5. DNS Server, Global Catalog and DSRM options configured.

4 Domain Controller Promotion

The **prerequisite check** completed successfully and the promotion process configured the server as a **domain controller**. The **DNS delegation warning** was expected because this isolated lab created a new forest with no existing parent DNS zone.

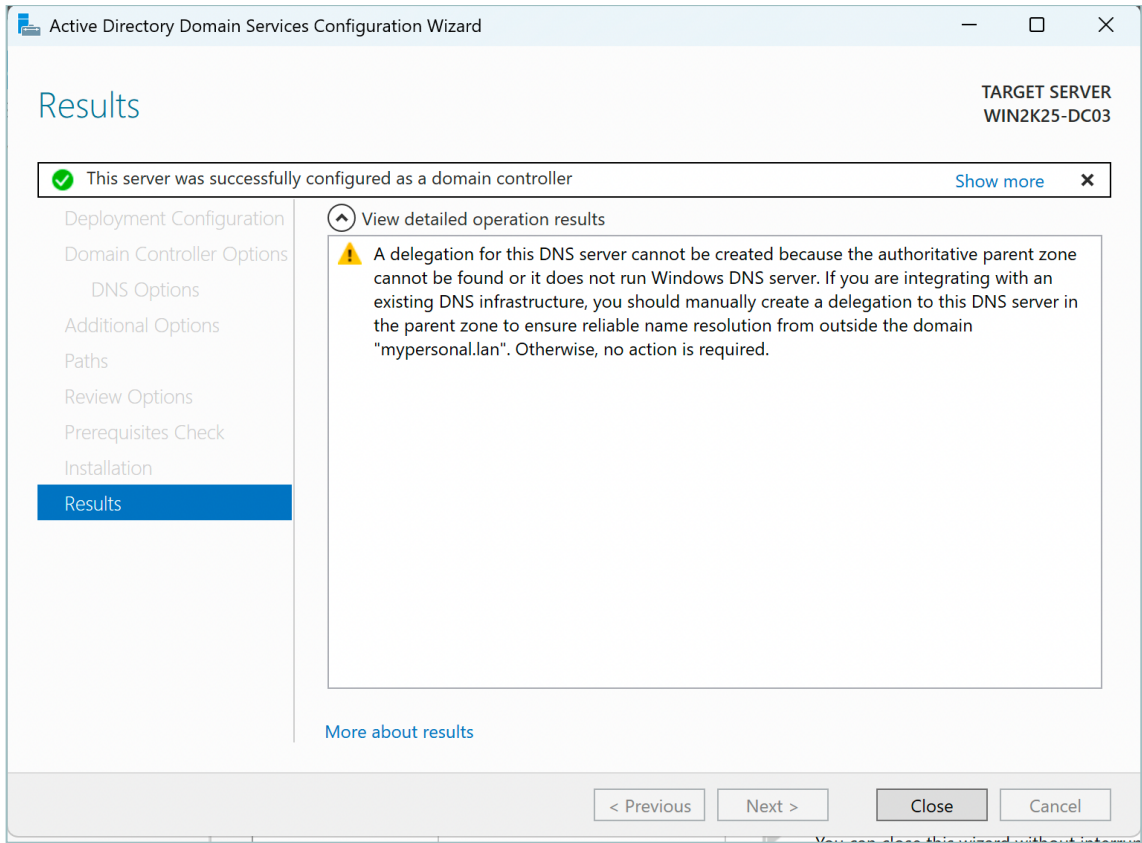


Figure 6. Domain-controller promotion completed successfully.

5 Post Promotion Verification

After promotion, the final **ipconfig /all** output confirmed the hostname **WIN2K25-DC03**, the **mypersonal.lan** DNS suffix, the static address **172.16.131.21**, gateway **172.16.131.2**, and DNS server **172.16.131.21**.

```
C:\Users\Administrator>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : WIN2K25-DC03
    Primary Dns Suffix . . . . . : mypersonal.lan
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : mypersonal.lan

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . : 
    Description . . . . . : vmxnet3 Ethernet Adapter
    Physical Address. . . . . : 00-0C-29-4F-6D-59
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 172.16.131.21(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.131.2
    DNS Servers . . . . . : 172.16.131.21
    NetBIOS over Tcpip. . . . . : Enabled

C:\Users\Administrator>
```

Figure 7. Final network configuration after domain-controller promotion.

6 Active Directory Management Access

Active Directory Administrative Center opened successfully and displayed the **mypersonal.ian** domain and its standard options. This confirmed that the directory could be accessed through the installed management tools.

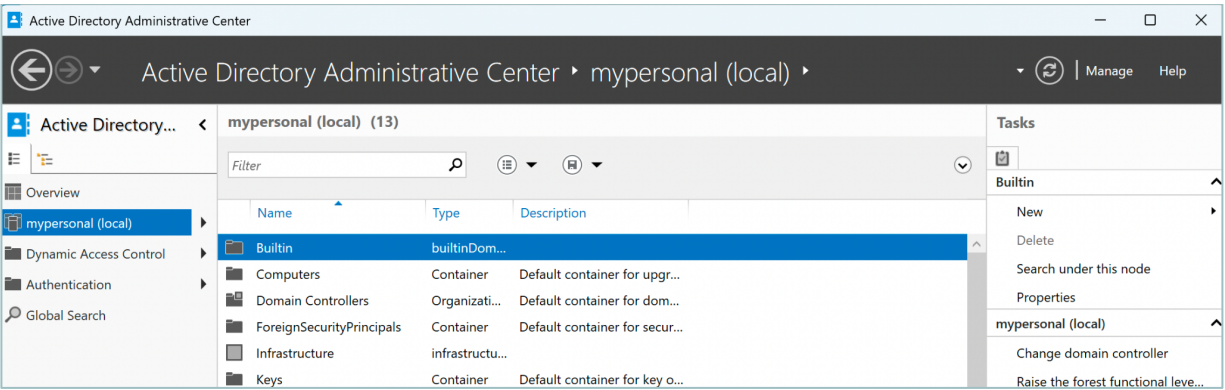


Figure 8. Active Directory Administrative Center displaying the new domain.

7 Reverse DNS and PTR Record

A **reverse lookup zone** was created for the **172.16.131.0/24** network. The **PTR record** linked **172.16.131.21** to **WIN2K25-DC03.mypersonal.lan**, which allows the server name to be found from its IP address.

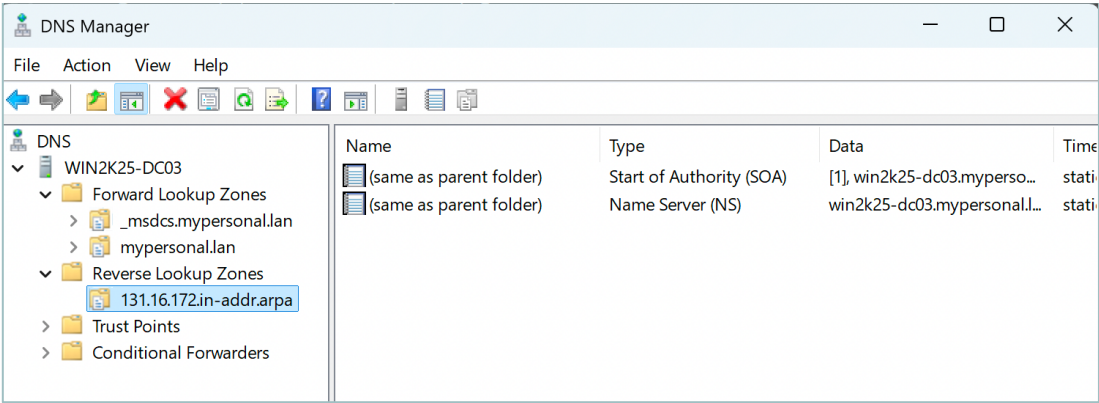


Figure 9. Reverse lookup zone visible in DNS Manager.

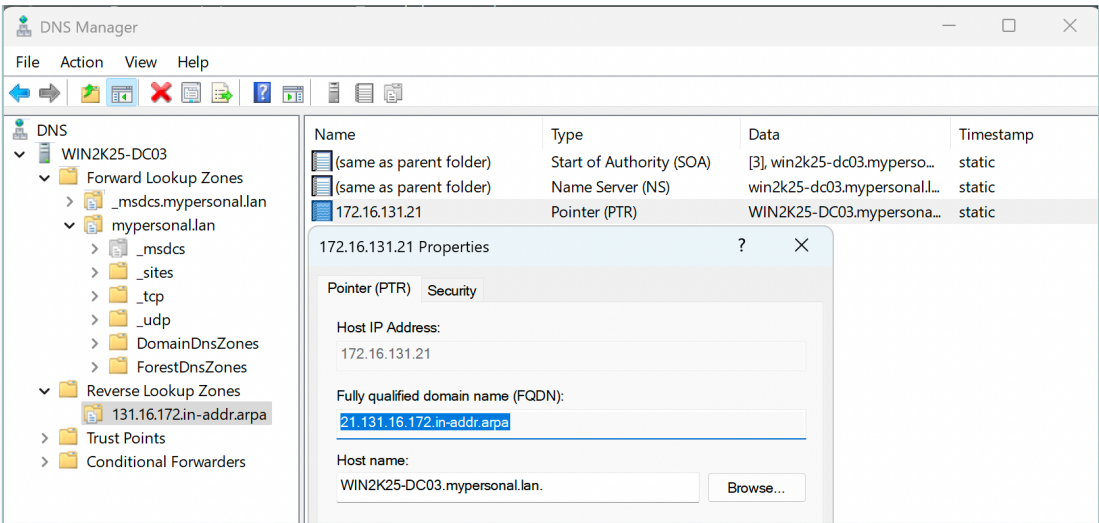
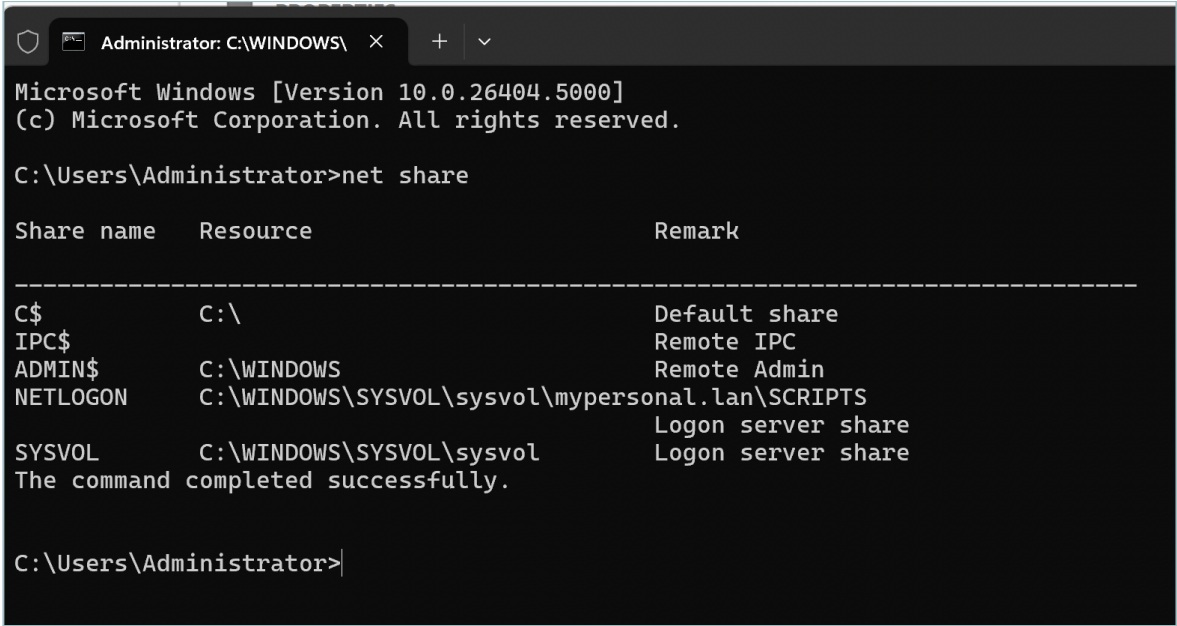


Figure 10. PTR properties linking 172.16.131.21 to the server hostname.

8 Verifying NETLOGON and SYSVOL

The **net share** command confirmed that the expected **NETLOGON** and **SYSVOL** shares were available. These shares support domain sign-in scripts, group policy, and other domain-controller functions. This check confirms share availability but does not, by itself, prove complete **domain-controller health**.



```
Administrator: C:\WINDOWS\
Microsoft Windows [Version 10.0.26404.5000]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>net share

Share name      Resource                                Remark
-----
C$              C:\                                    Default share
IPC$            C:\                                   Remote IPC
ADMIN$          C:\WINDOWS                           Remote Admin
NETLOGON        C:\WINDOWS\SYSVOL\sysvol\mypersonal.lan\SCRIPTS
Logon server share
SYSVOL          C:\WINDOWS\SYSVOL\sysvol             Logon server share
The command completed successfully.

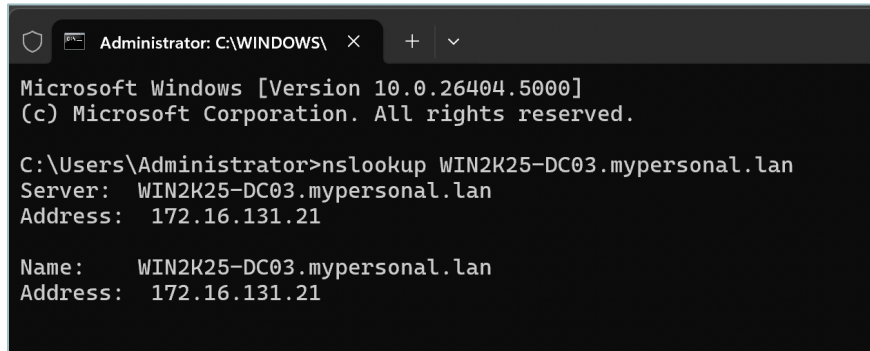
C:\Users\Administrator>
```

Figure 11. NETLOGON and SYSVOL listed by the net share command.

9 DNS Name Resolution

9.1 Forward lookup

The fully qualified server name **WIN2K25-DC03.mypersonal.lan** resolved to **172.16.131.21**, confirming the host **A record** and forward name resolution.

A screenshot of a Windows command prompt window titled "Administrator: C:\WINDOWS\". The window shows the output of the 'nslookup' command for the domain 'WIN2K25-DC03.mypersonal.lan'. The output displays the server name and its corresponding IP address, 172.16.131.21.

```
Microsoft Windows [Version 10.0.26404.5000]
(c) Microsoft Corporation. All rights reserved.

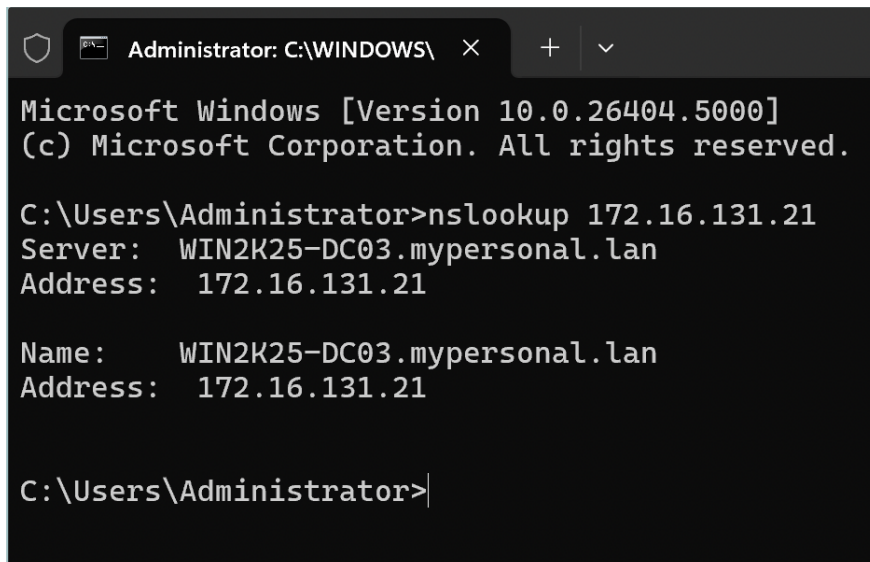
C:\Users\Administrator>nslookup WIN2K25-DC03.mypersonal.lan
Server:     WIN2K25-DC03.mypersonal.lan
Address:    172.16.131.21

Name:       WIN2K25-DC03.mypersonal.lan
Address:    172.16.131.21
```

Figure 12. Forward lookup resolving the server name to 172.16.131.21.

9.2 Reverse lookup

The address **172.16.131.21** resolved back to **WIN2K25-DC03.mypersonal.lan**, which confirms that the reverse zone and **PTR record** responded correctly.

A screenshot of a Windows command prompt window titled "Administrator: C:\WINDOWS\". The window shows the output of the 'nslookup' command for the IP address '172.16.131.21'. The output displays the IP address and its corresponding server name, WIN2K25-DC03.mypersonal.lan.

```
Microsoft Windows [Version 10.0.26404.5000]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>nslookup 172.16.131.21
Server:     WIN2K25-DC03.mypersonal.lan
Address:    172.16.131.21

Name:       WIN2K25-DC03.mypersonal.lan
Address:    172.16.131.21

C:\Users\Administrator>
```

Figure 13. Reverse lookup resolving the IP address to the server hostname.

10 Domain Controller Discovery and Health

The **LDAP SRV** lookup returned **WIN2K25-DC03.mypersonal.lan** on **TCP port 389** and linked the server to **172.16.131.21**. This record helps domain members locate an **Active Directory domain controller**.

```
C:\Users\Administrator>nslookup -type=SRV _ldap._tcp.dc._msdcs.mypersonal.lan
Server:  WIN2K25-DC03.mypersonal.lan
Address:  172.16.131.21

_ldap._tcp.dc._msdcs.mypersonal.lan    SRV service location:
      priority      = 0
      weight        = 100
      port          = 389
      svr hostname  = win2k25-dc03.mypersonal.lan
win2k25-dc03.mypersonal.lan    internet address = 172.16.131.21

C:\Users\Administrator>
```

Figure 14. LDAP SRV lookup identifying WIN2K25-DC03 as the domain controller.

Recorded dcdiag results

Diagnostic test	Recorded result
Connectivity	Passed
Advertising	Passed
SysVolCheck	Passed
NetLogons	Passed
Replications	Passed
Services	Passed
LocatorCheck	Passed
DFSREvent	Passed

11 Final Outcome and Key Learning

The deployment produced a working **Windows Server 2025 domain controller** for the isolated **mypersonal.ian** lab. The evidence confirms the server identity, static networking, **Active Directory** access, forward and reverse **DNS**, the **PTR record**, **NETLOGON** and **SYSVOL** availability, and successful forward, reverse, and **LDAP SRV** lookups.

Key learning

- A domain controller should have a stable hostname and static IP address before promotion.
- Active Directory depends on DNS records so computers can locate domain controllers and services.
- The DSRM password is a recovery password and is not used for normal domain Administrator sign-in.
- A reverse lookup zone is useful for troubleshooting but is not required for basic Active Directory operation.
- Diagnostic failures should be recorded accurately and investigated rather than presented as successful results.

Production considerations

This configuration was created for a **controlled lab**. A production deployment would require additional planning for domain naming, **redundancy**, **backups**, patching, **time synchronization**, **DNS forwarders**, monitoring, **least-privilege** administration, and recovery procedures. A **second domain controller** would normally be used to reduce dependence on a single server.